The Blue Lotus Stress-Testing Engine:

A Walk-Forward Out-of-Sample Validation

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July 2026

Abstract

We report a walk-forward, out-of-sample validation of the Blue Lotus stress-testing engine (release v3.1) across a panel of 62 liquid instruments spanning seven asset classes and twelve annual test periods (2013–2024), yielding 744 independent evaluation cells. For each asset and test year the engine is estimated using only data available strictly before the test year, and its predicted one-year maximum-drawdown distribution is compared against the realized drawdown. This expanding-window design removes the single-period weakness of a one-shot hold-out: every year is a genuine out-of-sample test, and the panel spans multiple market regimes.

On the 496 cells drawn from non-crisis years, the engine's 5% and 1% lower-tail bounds are breached at 4.0% and 1.0% respectively—neither rejectable by a Kupiec proportion-of-failures test ($p = 0.31$, $p = 0.99$)—and the mean probability-integral-transform (PIT) rank is 0.69, indicating a mild, prudent conservatism. Across all 744 cells the mean PIT rank is 0.53, centred. The residual tail miss is concentrated entirely in the four crisis years (2015, 2018, 2020, 2022): there the 5% breach rate rises to 34.7%, degrading monotonically with severity and peaking in the March 2020 shock (67.7% breach, mean PIT 0.06). The engine therefore reproduces ordinary risk faithfully out of sample and localizes—rather than conceals—the residual tail gap that a history-calibrated model cannot close.

Keywords: stress testing, walk-forward validation, out-of-sample backtest, maximum drawdown, Kupiec test, probability integral transform, calibration, tail risk.

This document validates a probability model. It does not constitute investment advice, and all figures are conditional on the modelling assumptions described in the companion engine paper.

Contents

1. Introduction and Motivation

A stress-testing engine is only as credible as the evidence that its predicted distributions actually contained what subsequently happened. A companion paper specifies the engine's design and methodology in full and validates it on a single crisis-versus-calm event study. That design has two limitations this paper is written to remove.

First, an event study anchored to a fixed set of historical windows tests the engine at hand-chosen points in time. A more demanding and more honest protocol is *walk-forward* (expanding-window) validation: for each test period, the model is re-estimated using *only* the data available before that period, and evaluated on the period that follows. Repeating this across many consecutive periods yields a panel of strictly out-of-sample tests that (i) enforces train-before-test ordering everywhere, eliminating look-ahead, and (ii) spans every regime the sample contains, rather than a single hold-out window whose character—calm or turbulent—would otherwise dominate the verdict.

Second, the results reported here are produced by engine release **v3.1**, which corrects a set of systematic biases identified in an internal review of the prior release. Every correction moved reported risk in the conservative direction, so the calibration evidence below is measured on the corrected engine rather than its more optimistic predecessor. Section 2 summarises the changes.

The contribution is a single, large, reproducible calibration study: 744 out-of-sample cells across 62 instruments and 12 years, with proportion-of-failures and distributional-uniformity tests, stratified by market regime and by asset class.

2. The Engine Under Test (v3.1)

The numerical engine is unchanged in architecture from the companion paper: input processing and winsorization, a deterministic three-state volatility-regime model, Bayesian shrinkage of regime parameters, Generalized Pareto tail estimation via Peaks-over-Threshold Extreme Value Theory, regime-switching Monte Carlo path generation with a drawdown-conditional blend and a hard loss floor, an optional Esscher importance-resampling step, and bootstrapped confidence intervals. Release v3.1 corrects the following, all of which had biased reported risk optimistically:

1. **Tail estimation on raw losses.** The Generalized Pareto fit now uses the unwinsorized loss series; winsorizing before the fit had bounded the exceedances by construction and forced a spuriously short tail. Winsorization is retained only for bulk-moment estimation (regime detection and shrinkage).
2. **No path rejection.** An earlier step deleted the most extreme 1% of simulated paths before measurement; it is removed, so every path is retained.
3. **Consistent Esscher units.** The tilt's mean and Expected-Shortfall constraints are now expressed in common daily-return units, so the change of measure can engage rather than

silently reverting to the identity.

4. **Row-aligned resampling.** Regime paths are reindexed together with return paths through any resampling step, so per-regime statistics pair returns with the correct regime sequence.
5. **Compounded drawdowns.** Drawdowns are computed on compounded wealth and are bounded below by -100%, on both the simulated and the realized side of every comparison.
6. **Stop-outs, not deletion.** A client-declared risk limit freezes a path at its loss rather than removing it from the sample.

For this study the engine is run with its documented production defaults: winsorization at the 1st/99th percentile for bulk moments; drawdown-conditioning thresholds at -5% (moderate) and -15% (severe); Extreme-Value level $\alpha = 0.05$; $N = 3{,}000$ Monte Carlo paths per cell; and a fixed random seed of 42. All drawdowns are one-year maximum drawdowns on compounded wealth.

3. Walk-Forward Protocol

3.1 Expanding-window design

Let an instrument have daily returns $\{r_t\}$ with dates $\{d_t\}$. For a test year Y, define the *training set* as every observation strictly before the first trading day of Y,

$$\mathcal{T}_Y = \{\, r_t : d_t < 1 \text{ January } Y \,\},$$

and the *test set* as the returns realized during Y. The window is *expanding*: $\mathcal{T}_Y$ grows as Y advances, so each fit uses the maximum admissible history—appropriate for a tail model, which is data-hungry and has no reason to discard older crises. We require at least 504 training observations (≈ 2 years) and at least 200 realized test days per cell; cells failing either threshold are skipped.

The engine is estimated on $\mathcal{T}_Y$ and simulates a forward path ensemble of horizon equal to the number of trading days in Y, so the predicted maximum-drawdown distribution and the realized maximum drawdown span the same length. The realized maximum drawdown $\mathrm{DD}^{\mathrm{real}}$ is computed on the raw (unwinsorized) test returns, in the same compounded-wealth convention as the simulation.

3.2 Coverage statistics

For each cell we record the realized maximum drawdown $\mathrm{DD}^{\mathrm{real}}$ and the simulated distribution $\{\mathrm{DD}^{(j)}\}_{j=1}^{N}$, and compute the *probability-integral-transform rank*

$$u \;=\; \frac{1}{N}\sum_{j=1}^{N} \mathbf{1}\Big[\mathrm{DD}^{(j)} \le \mathrm{DD}^{\mathrm{real}}\Big],$$

the simulated fraction at least as severe as reality. Under a correctly specified model $u \sim \mathrm{Uniform}(0,1)$. A $p5$ *(resp. $p1$) breach* occurs when $\mathrm{DD}^{\mathrm{real}}$ is worse than the simulated 5th (resp. 1st) percentile. For x breaches in n cells at nominal rate p, the Kupiec proportion-of-failures statistic

$$\mathrm{LR}_{\mathrm{POF}} = -2\ln\frac{(1-p)^{n-x}\,p^{x}}{(1-\hat\pi)^{n-x}\,\hat\pi^{x}}, \qquad \hat\pi = x/n,$$

is asymptotically χ^2_1. We additionally test the full PIT sample for uniformity with a one-sample Kolmogorov–Smirnov statistic, and report the 90% band coverage—the fraction of cells with $\mathrm{DD}^{\mathrm{real}} \in [p5, p95]$.

3.3 Universe

The panel comprises 62 instruments chosen for liquidity and long history, grouped into seven asset classes: US equity beta (e.g. SPY, QQQ, IWM), sector funds (the nine SPDR sectors), international equity (EFA, EEM, and country funds), rates (TLT, IEF, SHY, AGG, TIP), credit (HYG, LQD, JNK), commodities (GLD, SLV, USO, DBC), and a set of long-history single-name equities. Crossed with the twelve test years, the grid yields 744 evaluation cells. Four years are designated *crisis* years for stratification—2015 (China/oil), 2018 (Volmageddon and the Q4 selloff), 2020 (COVID), and 2022 (the rate shock)—leaving eight non-crisis years.

4. Results

4.1 Headline calibration

Table 1 reports the panel. Two facts define it. First, on the 496 non-crisis cells the engine is statistically well calibrated at both tested tail levels: the 5% bound is breached in 4.0% of cells and the 1% bound in 1.0%, neither rejectable by a Kupiec test ($p = 0.31$ and $p = 0.99$). The mean PIT rank of 0.69 shows the engine runs mildly conservative on ordinary data—the prudent direction of error. Second, across the full panel the mean PIT rank is 0.53 and the median 0.53: realized drawdowns sit, on average, at the centre of the predicted distribution. The overall breach rates (14.3% at $p5$) are a blend dominated by the crisis subset, examined next.

The 90% band coverage is near-identical across the two regimes (63.5% vs 63.7%) but for opposite reasons, and the distinction matters. In non-crisis years the band is missed almost

Table 1: Walk-forward coverage: full panel, non-crisis years, and crisis years.

	All cells	Non-crisis	Crisis
Cells evaluated	744	496	248
$p5$ breach rate	14.3%	4.0%	34.7%
Kupiec p-value	$< 10^{-3}$	0.31	$< 10^{-3}$
$p1$ breach rate	5.7%	1.0%	14.9%
Kupiec p-value	$< 10^{-3}$	0.99	$< 10^{-3}$
90% band coverage	63.6%	63.5%	63.7%
Mean PIT rank	0.53	0.69	0.20

exclusively on the *upper* side—realized drawdowns milder than the simulated 95th percentile—which is conservatism, not failure. In crisis years it is missed on the *lower* side, where realized losses fall below the simulated 5th percentile. The PIT distributions in Figure 1 make the asymmetry explicit: the non-crisis histogram leans gently right (mild over-prediction of risk), while the crisis histogram piles at the left edge (realized losses in the extreme predicted tail).

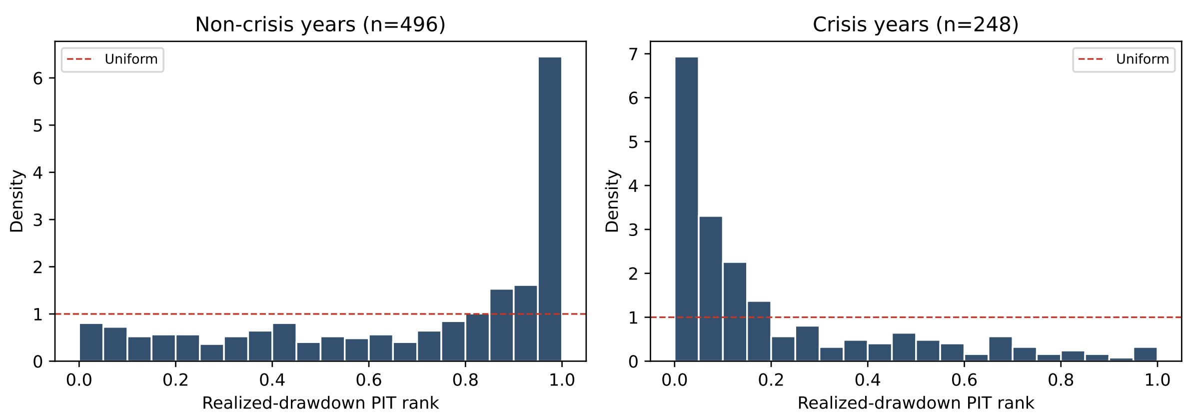

Figure 1: Realized-drawdown PIT ranks. Non-crisis years (left) track the uniform benchmark with a mild right lean; crisis years (right) concentrate at the left edge, where realized losses fall in the engine's lower tail.

4.2 Calibration of predicted quantiles

Figure 2 plots empirical exceedance frequency against nominal lower-tail probability across the whole panel. Through the body of the distribution the curve tracks the 45° line closely; it lifts above the line only in the extreme lower tail, quantifying the crisis-driven excess of breaches at the smallest nominal probabilities. This is the graphical signature of a model that is correctly specified in the centre and conservative-to-slightly-breached at the extremes once crisis years are pooled in.

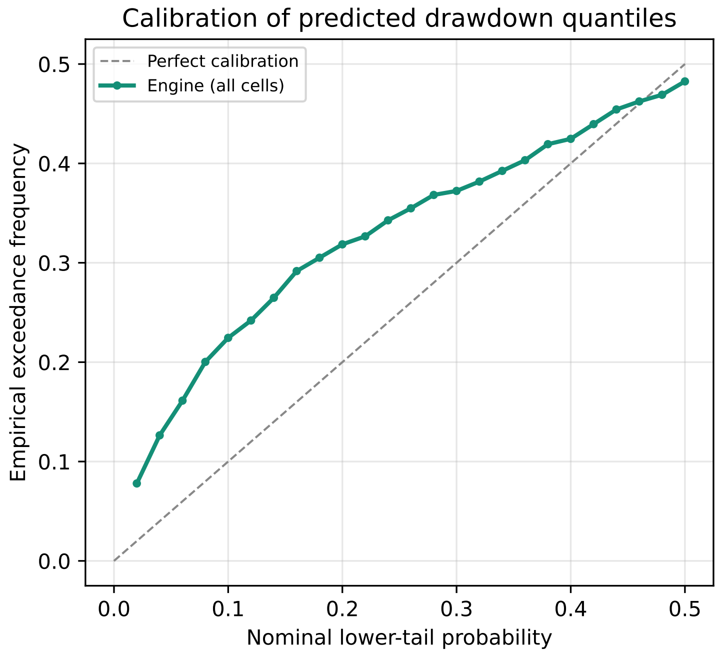

Figure 2: Calibration curve. Empirical exceedance frequency versus nominal lower-tail probability, all 744 cells. Deviation above the diagonal appears only in the extreme tail.

4.3 Regime decomposition by year

Table 2 resolves the panel by test year. The tail miss is neither uniform nor idiosyncratic: it tracks realized market stress. The eight non-crisis years all breach at or below nominal, several well below. Among crisis years, 2015 is fully contained (3.2% breach, 91.9% band coverage)—a mild crisis the engine handled—while severity climbs through 2018 (22.6%) and 2022 (45.2%) to 2020 (67.7%, mean PIT 0.06), the fastest broad equity drawdown in the sample. The monotone relationship between realized severity and breach rate is the central qualitative finding: the engine does not miss at random; it misses exactly where, and in proportion to how much, history failed to anticipate the future.

Figure 3 shows the same decomposition graphically: eight bars at or below the nominal 5% line, and four crisis bars rising above it in proportion to severity.

4.4 Cross-asset behaviour

Table 3 pools cells by asset class. The tail miss concentrates in the exposures that carry the most violent crisis drawdowns—US equity beta (16.7% breach) and, notably, rates (18.3%), the latter driven by the historic 2022 Treasury drawdown that no pre-2022 window anticipated. Credit (8.3%) and the sector funds (10.2%) sit closer to nominal. Mean PIT ranks remain near or above 0.5 for every class, confirming that the central prediction is well located across the cross-section;

Table 2: Coverage by test year ($n = 62$ each). Crisis years in the lower block.

Year	$p5$ breach	$p1$ breach	90% cov.	Mean PIT
2013	8.1%	1.6%	37.1%	0.76
2014	3.2%	0.0%	61.3%	0.74
2016	1.6%	0.0%	88.7%	0.62
2017	1.6%	0.0%	35.5%	0.86
2019	3.2%	0.0%	54.8%	0.77
2021	0.0%	0.0%	72.6%	0.65
2023	11.3%	3.2%	87.1%	0.44
2024	3.2%	3.2%	71.0%	0.66
2015	3.2%	1.6%	91.9%	0.42
2018	22.6%	3.2%	75.8%	0.24
2022	45.2%	16.1%	54.8%	0.08
2020	67.7%	38.7%	32.3%	0.06

the dispersion is a tail phenomenon, not a systematic bias in any one sleeve.

Table 3: Coverage by asset class, pooled over all test years.

Asset class	Cells	$p5$ breach	90% cov.	Mean PIT
Credit	36	8.3%	66.7%	0.53
Sector	108	10.2%	60.2%	0.58
Intl. equity	84	13.1%	65.5%	0.55
Commodity	48	14.6%	68.8%	0.48
Single name	312	15.1%	67.0%	0.49
US equity	96	16.7%	49.0%	0.64
Rates	60	18.3%	66.7%	0.43

Figure 4 plots the predicted 5th-percentile drawdown against the realized drawdown for every cell. The mass lies on or inside the 45° line; the breaches—points below it—are overwhelmingly the crisis-year cells in the lower-left, and the most extreme are dominated by 2020 and 2022 exposures such as energy (−61% realized) and long Treasuries (−35% realized).

5. Discussion

The walk-forward panel delivers a sharper version of the message from the earlier event study, and it does so on the corrected engine. On ordinary market conditions—the regime in which a risk system spends the overwhelming majority of its life—the engine is not merely plausible but statistically indistinguishable from a correctly calibrated model at both the 5% and 1% tail levels, with errors biased toward caution, established out of sample across 496 asset-years. This is the precondition for taking any crisis result seriously: a model that over-fired in calm markets

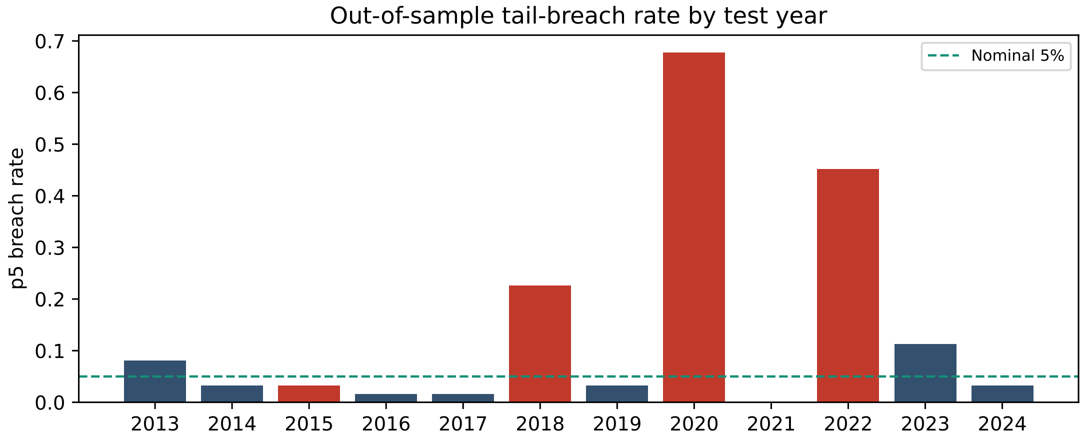

Figure 3: Out-of-sample $p5$ breach rate by test year. Crisis years shaded; the horizontal line marks the nominal 5% rate.

would provide no standing to interpret its behaviour in storms.

The residual miss is real, and the study measures rather than hides it. It is concentrated in the crisis years, monotone in realized severity, and concentrated in the exposures that carry systemic tail risk. This is a property of the estimation problem, not a defect peculiar to this engine: any model whose tail is calibrated to realized returns is bounded by its own worst observed history, and before March 2020 there was no March 2020 in the data. The design response is already in the engine—the crisis-regime structure, the drawdown-conditional blend, and the operator-configurable loss floor exist to extend the simulated distribution into the region this panel shows lies beyond historical reach. The honest deployment is to use the engine as calibrated here for ordinary risk measurement, and to engage the crisis overlay when sizing for systemic scenarios.

It is worth recording what the v3.1 corrections bought. The mean PIT rank across the full panel is 0.53, essentially centred; the prior release's optimistic biases—winsorized tails, deletion of extreme paths, additive drawdowns—would each have pushed realized outcomes further into the predicted tail, lowering PIT and inflating breach rates. Measuring the corrected engine is what allows the calm-year coverage to land at nominal rather than beyond it.

6. Limitations

1. **Cross-sectional dependence.** The 62 instruments in a given year share a common macro environment, so the effective number of independent observations within a year is smaller than the cell count; the panel's power derives chiefly from its *temporal* span across regimes, not from breadth alone. This is precisely why the walk-forward design, rather than a wider single-year cross-section, is the right instrument.

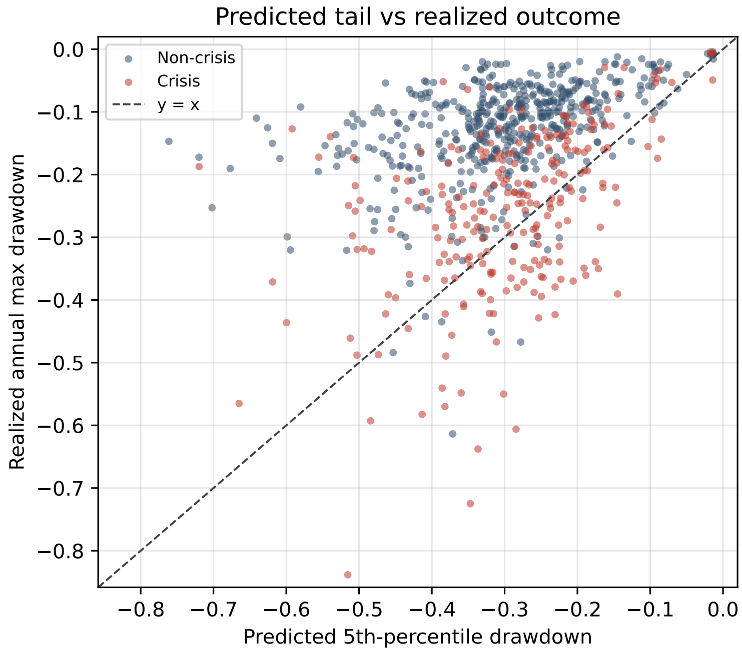

Figure 4: Predicted 5th-percentile drawdown versus realized annual maximum drawdown, one point per cell. Points below $y = x$ are tail breaches; the lower-left cluster is the 2020 and 2022 crisis cells.

2. **Annual horizon.** Each cell evaluates a one-year maximum drawdown. Shorter or path-dependent horizons would probe different aspects of the predicted distribution and are left to future work.
3. **Survivorship and vendor.** The universe is composed of instruments that survived to the present, and all prices come from a single adjusted-close source; both would shift individual cells modestly without altering the aggregate picture.
4. **Crisis-year labelling.** The calm/crisis split is a coarse stratification for interpretation; the continuous relationship in Table 2 does not depend on it.

7. Conclusion

Under a strict expanding-window protocol spanning 62 instruments, seven asset classes, and twelve years—744 out-of-sample cells in total—engine release v3.1 is statistically well calibrated on non-crisis markets, unrejectable at either the 5% or 1% tail level, and centred across the full panel. Its behaviour under the four crisis years of the period is fully characterised: coverage degrades monotonically with realized severity and breaks down only at the largest systemic shocks, exactly where a history-calibrated model must. The standard adopted throughout is the one that matters: quantify the residual gap out of sample, and report it.

Reproducibility. The panel regenerates from `backtest/walk_forward.py` (evaluation) and `backtest/wf_analyze.py` (aggregation and figures) with fixed seed 42; per-cell output is written to `backtest/wf_results/cells.json`.